Classification Trees

Tasks (Lab 6):

1. Classification tree.

   Consider the dataset BreastCancer (`sklearn.datasets.load_breast_cancer`).

   (a) Fit a classification tree. Analyze how different parameters (`criterion`, `splitter`, `max_depth`, `min_samples_split`) affect the structure, depth, and size of the tree.

   (b) Draw the fitted tree and briefly interpret the first splits.

   (c) Prune the default tree using the cost-complexity criterion. Compare the pruned and unpruned trees in terms of size and predictive performance.

2. Bagging.

   Implement your own version of the bagging algorithm. You may use available implementations of decision trees as base learners. Compare bagging with a single tree and comment on why bagging may improve stability and predictive performance.

3. Compare the following methods on 3 selected datasets:

   - Bagging,
   - Single tree,
   - Random Forest.

   Use default parameters. Compare the methods using the same train/test splits and report the results in a clear table. Briefly comment on the differences between the methods.